

Aniket Tayade

Machine Learning Engineer

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| github.com/tayade-aniket | Portfolio

Professional Summary

Machine Learning Engineer with 1.5+ years of experience designing and shipping end-to-end ML, deep learning, computer vision, and LLM systems in Python, spanning data engineering, model training, evaluation, and production deployment. Skilled in PyTorch, Hugging Face Transformers, QLoRA fine-tuning, federated learning, retrieval-augmented generation (RAG), and ONNX inference optimization. Published researcher in on-device LLM personalization and LLM faithfulness/compliance guardrails. Seeking Machine Learning Engineer opportunities internationally.

Technical Skills

Languages: Python, SQL, JavaScript

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers

LLM & GenAI: LangChain, RAG, QLoRA/PEFT, Prompt Engineering, NLI-based Faithfulness Scoring, Groq LLM

Computer Vision: YOLO, OpenCV, Tesseract OCR, Image Preprocessing, Object/Text Detection

MLOps & Deployment: FastAPI, Streamlit, Docker, ONNX/ONNX Runtime, Model Export & Inference Optimization, Git/GitHub, CI-ready Pipelines

Federated & Edge AI: Federated Learning, Federated Averaging (FedAvg), Privacy-Preserving AI, On-Device Inference

Data Engineering: ETL Pipelines, Pandas, NumPy, Data Validation, SQL Query Optimization, MySQL, MongoDB, SQL Server

Professional Experience

Junior Data Scientist

Feb 2025 – Present

NextHike IT Solutions, Remote

- Built and validated supervised ML models in Python/Scikit-learn for predictive analytics, covering preprocessing, feature engineering, and statistical evaluation.
- Engineered ETL pipelines and optimized SQL queries, reducing data processing time by 40% while improving pipeline reliability and ML data readiness.
- Processed large-scale datasets with Pandas, NumPy, and SQL to produce analysis-ready data for downstream modeling.
- Delivered Python and Power BI reporting solutions, improving stakeholder reporting efficiency by 35%.

Python Programming Instructor

Jan 2024 – Dec 2024

Vision Computers Academy, Pune, India

- Trained 100+ students in Python, SQL, object-oriented programming, data structures, and algorithms through hands-on, project-based coursework.

AI/ML Projects

LLM Hallucination & Compliance Guardrail System

GitHub

Python, FastAPI, ChromaDB (RAG), DeBERTa-v3 (NLI), Streamlit

- Designed a production-style guardrail layer that intercepts LLM outputs, retrieves grounding context via a ChromaDB RAG pipeline, and scores response faithfulness with a DeBERTa-v3 NLI judge, addressing hallucination and regulatory-compliance (GDPR/PII) risk in enterprise LLM deployments.
- Built a risk-tiered routing engine (pass/warn/escalate/block) with domain-specific thresholds and a full audit trail, targeting greater than 85% hallucination recall at under 500ms guardrail overhead.

Privacy-Preserving On-Device LLM Personalization (Federated QLoRA)

GitHub

Python, PyTorch, QLoRA, Transformers, Federated Learning, ONNX

- Built a privacy-first framework combining 4-bit QLoRA with Federated Averaging (FedAvg) to personalize LLMs on-device without transferring raw user data, cutting trainable parameters by over 95% versus full fine-tuning.
- Designed an ONNX Runtime export/deployment pipeline for low-latency on-device inference; formed the basis of a research paper presented at INSIGHT 2026.

Medical Report OCR Pipeline

GitHub

Python, YOLO, Tesseract OCR, OpenCV

- Built an end-to-end computer vision pipeline combining YOLO-based text detection with Tesseract OCR to extract structured data from unstructured medical lab report layouts, auto-exporting to CSV/JSON for downstream analytics.

Education

M.Sc. Computer Science (in progress) – CGPA 9.07/10

Jul 2025 – May 2027

MAEER's MIT Arts, Commerce and Science College, Pune, India

B.Voc. Software Development – CGPA 7.60/10 (equivalent to Bachelor's degree)

Jun 2016 – May 2019

Sant Gadge Baba Amravati University, Amravati, India

Publications & Research

- *Privacy-Preserving On-Device LLM Personalization via Federated QLoRA* – presented at INSIGHT 2026 (IJIRT / MIT ACSC, Pune)
- *Artificial Intelligence for Fraud Detection and Market Manipulation* – published in IJARPR, Nov 2025

Certifications

Post Graduate Program in Data Science & AI (Digicrome Academy) | Model Context Protocols: Advanced Topics (Anthropic) | AI Fluency: Framework & Foundations (Anthropic) | Generative AI Studio (Google Cloud)